

Distributed Systems

Workshop 2: Communication (Messaging and Message Queuing)

Semester II 2026
Aug 27, 2026
Prof. Francisco Hidrobo

Workshop Activities

Part 1: RMI

1. Review the example codes, RMI for server and client . Test programs:
 - 1.1. Same machine
 - 1.2. On different hosts (Client and server running on different hosts)
2. Create an application to implement a Distributed “Matrix Manager”. Client read (o generate) matrices and operations (add, sub, prod); and server compute and return result as matrix (use **numpy** for matrices).
 - 2.1. Write a Python version
 - 2.2. Test the program locally and using two different machines

Part 2: Publisher-Subscriber

3. Review the example codes publisher and subscriber. Test programs:
 - 3.1. Same machine
 - 3.2. On different hosts
 - 3.3. with more than one subscriber
4. Create a version (python) with multiples publishers and multiple subscribers. Each publishers offers different services, and subscribers can be subscribe to more than one publishers.

Part 3: Pipeline comunicacion

5. Review the example codes pipeline (Source -Worker). Test programs:
 - 5.1. Same machine
 - 5.2. On different hosts
6. Create a version (python) with Source → broker → Worker pipeline with a single input and output (in broker). Run with multiplles sources and multiples workers

Report

You should submit a report (same format used in workshop 1) with code and results (screenshots). Only parts 2, 4 and 6